

Aafiya Hussain

+1 (540) 934-8040 aafiyahussain@vt.edu [in LinkedIn](#) [on GitHub](#) [Google Scholar](#) [Website](#)

EDUCATION

Virginia Tech

Master of Science in Computer Science (Fully funded - **\$62,705/year** Scholarship), **GPA: 4.0/4.0**

Blacksburg, Virginia

Aug 2024 - May 2026

- Graduate Teaching Assistant for CS5704, 5244, 5814, 1114 (Fall 24 – Spring 26).

K.J. Somaiya College of Engineering

B.Tech in Computer Engineering, **GPA: 9.44/10**

Mumbai, India

Jul 2018 - Jul 2022

EXPERIENCE

Robot Toolworx

Machine Learning Engineer

Jun 2026 - Present

Rutherford, NJ, USA

- Building agentic workflows to assist non-expert aerospace, maritime, and automotive technicians using multimodal knowledge graphs for information retrieval.

Nokia Bell Labs

Computer Vision Intern (Industrial Metaverse)

Jun 2025 - Aug 2025

Murray Hill, NJ, USA

- Led development of an end-to-end multimodal LLM and multi-view stereo pipeline to automate querying of complex 3D scenes by building structured scene embeddings from RGB images, **reducing manual annotation effort by ~95%**.
- Built an interactive visualization layer with 3D Gaussian splats, enabling users to query scenes through a chat interface.
- Improved performance over diverse lighting conditions, heterogeneous sensors, and challenging occlusions (**10+ cases**), ensuring system robustness across diverse data-collection pipelines (**accuracy variance < 5%**).
- Enhanced the 3D Gaussian Splat reconstruction framework using accurate ChArUco board calibration, yielding **~25% higher geometric fidelity and ~18% better depth precision**.

Madan Bhandari University of Science and Technology

Research Scholar

Jun 2023 - Jul 2024

Bagmati, Nepal

- Fine-tuned Whisper models for automatic speech recognition (ASR) of **13+ Indic languages** to caption video-based explanations of undergraduate computer science coursework, eliminating manual translation effort.
- Integrated the ASR module into the University's learning management system, **improving student engagement by ~60%**.

Deloitte

Software Development Engineer (Analyst)

Nov 2022 - Jun 2023

Mumbai, MH, India

- Led the development of an automated fraud detection system processing **50K+ daily transactions**, achieving **94% accuracy** and flagging suspicious transfers for manual review.
- Deployed production-ready ensemble models to score client risk from financial profiles, achieving **92% accuracy** and strengthening end-to-end decision-making in compliance workflows.

SELECTED PUBLICATIONS

- A. Hussain**, G. Srivastava, A.M. Ishmam, and C. Thomas. "SoundBreak: A Systematic Study of Audio-Only Adversarial Attacks on Trimodal Models." [arXiv](#) [↗](#) . **ACL 2026 Main Conference** (19% acceptance).
- G. Srivastava, **A. Hussain**, C. Wang, Y. Celine and X. Wang. "effGen: Enabling Small Language Models as Capable Autonomous Agents." Framework available at [effGen.org](#) [↗](#) | [arXiv](#) [↗](#) . **ICML 2026** (26.6% acceptance)
- G. Srivastava, **A. Hussain**, Z. Bi, S. Roy, P. Pitre, M. Lu, M. Ziyadi, and X. Wang. "BeyondBench: Benchmark-Free Evaluation of Reasoning in Language Models." [OpenReview](#) [↗](#) | [Leaderboard](#) [↗](#) . **ICLR 2026** (28% acceptance).
- G. Srivastava, **A. Hussain**, S. Srinivasan, and X. Wang. "Do LLMs Overthink Basic Math Reasoning? Benchmarking the Accuracy-Efficiency Tradeoff in Language Models" [arXiv](#) [↗](#) . **ACL 2026 Findings** (18% Findings acceptance)
- P. Zhou, A. Asgarov, **A. Hussain**, W. Park, A. Paudyal + 7 authors. "Designing Multi-Robot Ground Video Sensemaking with Public Safety Professionals." [arXiv](#) [↗](#) . **CHI 2026** (25.3% acceptance).

*Complete list of publications - [Google Scholar](#) [↗](#) , citations: 204

SELECTED PROJECTS

LLMThinkBench [Python](#) | [vLLM](#) | [PyTorch](#) | [Transformers](#)

[GitHub](#) [↗](#) | [PyPI](#) [↗](#) | [Leaderboard](#) [↗](#)

- Developed a **dynamic math reasoning evaluation framework** to fairly measure LLMs' innate reasoning ability without benchmark contamination, covering **14 arithmetic tasks** with dual-backend inference system and multi-GPU parallelism.
- Engineered a production-ready PyPI package (**7.07k+ downloads**) with **17 configurable parameters**, **140+ regex-based answer extraction methods**; released leaderboard with **results from 55+ LLMs**.
- Proposed a novel **Overthinking Score** metric balancing accuracy with token efficiency using F1-harmonic mean.

TECHNICAL SKILLS

Programming Languages

Python, Java, JavaScript

Libraries & Frameworks

PyTorch, TensorFlow, Transformers, vLLM, scikit-learn, OpenCV, LangChain

Technologies

Flask, MySQL

Tools

AWS, FastAPI, Docker